

# Digital Image Processing

## Practical Assignment 01

Platform: MATLAB/Python

1. Write a MATLAB program to:  
Load grayscale and RGB image using `imread`.  
Display the image using `imshow`.  
Use `size()` to display the dimensions (height, width, channels).
2. Write a MATLAB program to:  
Convert the loaded color image to `grayscale` using `rgb2gray`.  
Display both original and grayscale images in a single figure using subplot.
3. Write a MATLAB program to:  
Convert an RGB image to `HSV` using `rgb2hsv`.  
Display the `Hue`, `Saturation`, and `Value` components separately.
4. Write a MATLAB program to:  
Save the grayscale image, say `cameraman.tif` to your computer using `imwrite`.
5. Compare the appearance of the same image in `RGB`, `Grayscale`, and `HSV` formats. Write your observations about how details differ in each representation.